

Data Visualization and Exploration
Assignment 2: NYC Taxi Trip Analysis

COMS4060A/7056A

School of Computer Science & Applied Mathematics
University of the Witwatersrand

Bonolo Lebepe	2548042
Tebogo Seopa	2563912
Adivhaho Nevondo	2580619
Simphiwe Rasenyalo	2457104

Introduction

The data is the training sample of the NYC Taxi and Limousine Commission trip records released for the Kaggle *NYC Taxi Trip Duration* competition [4]. It contains 1,458,644 trips between 2016-01-01 and 2016-06-30 with 11 columns: `id`, `vendor_id`, `pickup_datetime`, `dropoff_datetime`, `passenger_count`, `pickup_longitude`, `pickup_latitude`, `dropoff_longitude`, `dropoff_latitude`, `store_and_fwd_flag` and `trip_duration` (seconds). No distance or speed column is provided, so both are derived.

Conventions used throughout. Only the training file is used. All analyses after Section 1 use the cleaned data (1,442,031 trips), and any further filter is stated in the section that applies it. Times are pickup times in New York local time unless stated otherwise. Distances are great-circle (haversine) distances from the function supplied with the assignment. The assignment labels its reference points as "(long, lat)", but the values are latitude first, so we read them as (latitude, longitude): for example, JFK is at latitude 40.647929, longitude -73.777813. Every random step uses seed 42. Sections 1–6 answer Questions 1–6 of the assignment in order.

1 Data Cleaning

1.1 Initial diagnostics

Table 1: Summary statistics of the raw (uncleaned) data. `distance_km` is the haversine distance between pickup and drop-off.

	`trip_duration` (s)	`distance_km`	`speed_kmh`
count	1,458,644	1,458,644	1,458,644
mean	959.49	3.44	14.41
std	5,237.43	4.29	14.97
min	1.00	0.00	0.00
1 %	87.00	0.14	1.39
50 %	662.00	2.09	12.78
99 %	3,440.00	20.77	40.72
max	3,526,282	1,240.13	9,269.01

The maximum trip duration is **40.8 days**, the maximum distance is **1,240 km** (roughly the distance to Chicago) and the maximum implied speed is **9,269 km/h** (Table 1). None of these can be real taxi trips inside New York City. Row-level checks found:

- `trip_duration` ≤ 0: 0 rows (the minimum is 1 s);
- `trip_duration` > 6 h: 2,061 rows;
- zero-distance trips: 5,897 rows;
- `speed_kmh` > 100: 175 rows;
- `speed_kmh` < 1: 13,241 rows;
- pickup outside the NYC bounding box: 256 rows;
- `passenger_count`: 60 trips with 0 passengers, and 5 trips with 7, 8 or 9 (above the legal capacity of a standard cab).

1.2 Cleaning rules and justification

Five filters were applied, each motivated by a physical or operational argument (Table 2).

1.3 Retention and effect

Table 3 gives the share of rows each filter keeps on its own. Together they keep **1,442,031 rows (98.86%)**, removing 16,613 rows (1.14%).

The cleaned distribution is bounded and realistic (Table 4). The median duration is 664 s (11.1 min), the median distance 2.11 km and the median speed 12.85 km/h, and the standard deviation of duration

Table 2: Cleaning rules and their justification.

Rule	Threshold	Justification
Duration	$30\,\text{s} \leq$ `trip_duration` $\leq 6\,\text{h}$	Trips under 30 s are meter toggles, cancellations or clock errors. NYC is about 60 km across, so even the longest realistic trips take a few hours; multi-day values are meters left running.
Speed	$1 \leq$ `speed_kmh` ≤ 100	The 1 km/h floor removes stationary GPS records (waiting, signal loss). No *average* speed on NYC roads can exceed 100 km/h.
Geographic box	lat $\in [40.5, 41.0]$, lon $\in [-74.3, -73.7]$	Covers all five boroughs and the JFK, LaGuardia and Newark airports. Points outside are GPS glitches such as $(0, 0)$ or mistyped values.
Non-zero distance	`distance_km` > 0	A trip with zero distance but non-zero duration is a meter movement or GPS failure and has no spatial content.
Passengers	$1 \leq$ `passenger_count` ≤ 6	A count of 0 is a data-entry error; more than 6 exceeds legal cab capacity.

Table 3: Rows kept by each filter on its own, and by all filters together.

Filter	Rows kept	Share
Duration 30 s – 6 h	1,451,801	99.53 %
Speed 1 – 100 km/h	1,445,228	99.08 %
Coordinates in NYC box	1,457,646	99.93 %
Distance > 0	1,452,747	99.60 %
Passenger count 1–6	1,458,579	100.00 %
All filters together	**1,442,031**	**98.86 %**

Table 4: Summary statistics of the cleaned data.

	`trip_duration` (s)	`distance_km`	`speed_kmh`
count	1,442,031	1,442,031	1,442,031
mean	838.80	3.45	14.48
std	656.25	3.88	7.62
min	30.00	0.009	1.00
25 %	400.00	1.251	9.22
50 %	664.00	2.112	12.85
75 %	1,074.00	3.896	17.88
max	21,411.00	47.735	99.47

falls from 5,237 s to 656 s. Removing these rows is justified because they are *not* a random sample of the data: they sit in the tails, so they would bias every mean, plot and model downstream, and since they are not real taxi trips they carry no information about travel demand.

2 Feature Generation

`distance_km` and `speed_kmh` are computed *before* cleaning, so that the speed filter can be applied to every row. The time features are computed *after* cleaning (Table 5).

3 Time-Based Analysis

3.1 Most popular day of the week

Friday is the busiest day, with 220,968 trips (15.32%). Monday is the quietest (185,308 trips, 12.85%); see Table 6 and Figure 1.

Table 5: Generated features.

Feature	Definition	Purpose
`distance_km`	haversine(pickup, drop-off)	The data has no distance column, and every speed and spatial analysis needs one. The straight-line distance underestimates road distance in Manhattan but is consistent and easy to compute.
`speed_kmh`	distance (km) divided by duration (h)	Average speed of the trip; used by the speed filter and in Section 3.5.
`pickup_dow`	day of week, Monday $= 0$	Grouping by day of week.
`day_name`	day name	Readable labels for tables and plots.
`pickup_hour`	hour of pickup, 0–23	Hourly analysis in Sections 3.2–3.5.
`is_weekend`	`pickup_dow` ≥ 5	Weekday/weekend split in Section 3.3.

Table 6: Trips by day of week (cleaned data).

Day	Trips	Share (%)
Monday	185,308	12.85
Tuesday	200,459	13.90
Wednesday	207,871	14.42
Thursday	216,049	14.98
Friday	220,968	15.32
Saturday	218,393	15.15
Sunday	192,983	13.38

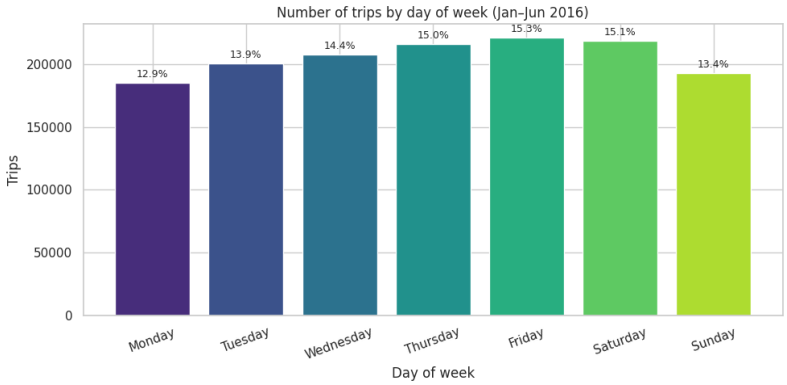


Figure 1: Trips by day of week, with percentage shares.

Demand rises steadily from Monday to Friday, stays high on Saturday and drops on Sunday. Friday combines the end of the working week (commuting and after-work socialising) with the start of the weekend (nightlife and visitors). Sunday loses both the commuters and the business travellers.

3.2 Peak hour on each day

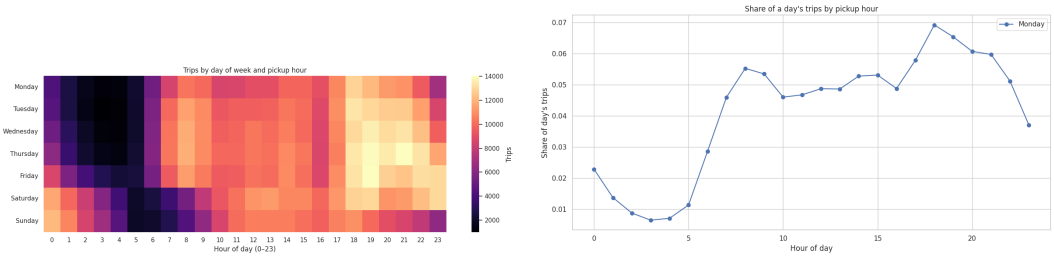
Table 7 gives the busiest pickup hour on each day. Figure 2 shows the day-by-hour heatmap and each day's hourly profile.

Observations and possible explanations.

- **Monday to Wednesday** peak at **18:00–19:00**, the end-of-work commute, when taxis are used for the trip home and for after-work plans.
- **Thursday and Friday** peak later or higher (**21:00** and **19:00**, with the two largest peak counts). As the week ends, after-work demand merges into evening and nightlife demand.
- **Saturday** peaks at **23:00** and **Sunday** at **00:00**: late-night demand from bars, restaurants and clubs that runs past midnight.

Table 7: Busiest pickup hour on each day and its trip count.

Day	Peak hour	Peak trips
Monday	18:00	12,829
Tuesday	18:00	13,328
Wednesday	19:00	13,573
Thursday	21:00	14,019
Friday	19:00	14,030
Saturday	23:00	13,051
Sunday	00:00	12,143



(a) Trips by day and hour.

(b) Share of each day's trips by hour.

Figure 2: Hourly demand on each day of the week.

- The weekday *morning* peak at 08:00 is visible but always smaller than the evening peak, consistent with many commuters taking the subway in the morning and a taxi in the evening.
- Every day reaches its minimum between 03:00 and 05:00. The trough comes later at weekends (05:00, against 03:00–04:00 on weekdays) because late-night demand runs on into the small hours.

3.3 Weekdays versus weekends

The weekday peak is at **19:00 (512 trips per day on average)** and the weekend peak at **00:00 (457 trips per day)**. Figure 3 shows the full hourly curves.

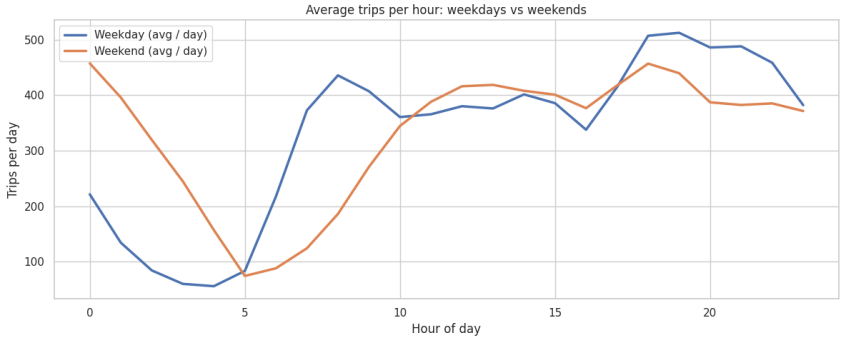


Figure 3: Average trips per hour: weekdays versus weekends.

Differences and explanations.

- **Weekdays** have two peaks: a small morning peak around 08:00 and a large evening peak at 18:00–19:00. This is the commuting signature of fixed work schedules.
- **Weekends** have *no* morning peak. Demand builds through the afternoon and peaks at midnight, following leisure schedules: dinner, nightlife, tourism and shopping.
- The largest difference is in the small hours. At 03:00–04:00 weekdays are at their daily minimum (about 116 trips per day over those two hours), while weekends still carry about 402, because nightlife demand is still active.

3.4 Major holidays

The 2016 dates are **Martin Luther King Jr. Day, Monday 18 January**; **Valentine's Day, Sunday 14 February**; **St. Patrick's Day, Thursday 17 March**; **Easter Sunday, 27 March**; and **Memorial Day, Monday 30 May**. Each holiday is compared with the average of the other 25 days of the same weekday, so the day-of-week effect is removed. Figure 4 shows each holiday's hourly profile, and Table 8 the changes in trip counts.

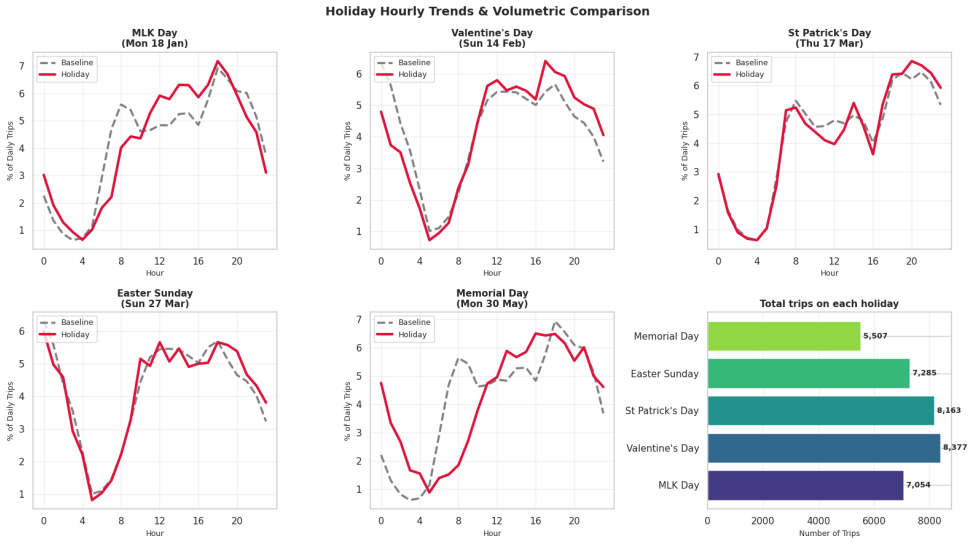


Figure 4: Hourly share of the day's trips on each 2016 holiday against the same-weekday baseline, and the total trips on each holiday.

Table 8: Holiday trip counts relative to the average of the other days of the same weekday.

Holiday	Pattern against baseline	Interpretation
MLK Day (Mon 18 Jan)	Total -1%. 07:00–09:59 at 67% of a normal Monday; 10:00–15:59 up 14%; peak stays at 18:00.	Fewer commuters, more daytime leisure trips.
Valentine's Day (Sun 14 Feb)	Total $+13\%$. 16:00–23:59 up 26–34%; 00:00–06:59 down 17%; peak moves from 00:00 to 17:00.	Dinner and date-night trips.
St. Patrick's Day (Thu 17 Mar)	Total -2%. Daytime slightly below normal; 22:00–02:59 up 12% on other Thursdays; peak at 20:00 instead of 21:00.	Parade and pub evening; a modest, late effect.
Easter Sunday (Sun 27 Mar)	Total -2%; every time block within 10% of a normal Sunday.	Little change in taxi use.
Memorial Day (Mon 30 May)	Total -23%. 07:00–09:59 at 30% of normal; later blocks about 20% lower; 00:00–06:59 up 29%; peak moves from 18:00 to 16:00.	Long weekend: many residents away, and late Sunday-night trips carry over.

Only Memorial Day (-23%) and Valentine's Day ($+13\%$) change the daily total by more than 2%. The two Monday holidays lose much of the morning commute, MLK Day falling to 67% and Memorial Day to 30% of a normal Monday's morning trips. St. Patrick's Day and Easter look almost like ordinary days, apart from a small late-evening rise on St. Patrick's. Holidays therefore change *when* people travel more than *how much*, with Memorial Day as the exception. The baselines are averages over many days, which themselves include other holidays and bad-weather days.

3.5 Average speed by hour of day

Findings (Table 9, Figure 5).

Table 9: Mean and median speed (km/h) by pickup hour.

Hour	Mean	Median	Hour	Mean	Median
0	17.76	16.09	12	12.30	10.81
1	18.53	17.03	13	12.49	11.07
2	18.96	17.60	14	12.27	10.93
3	20.04	18.58	**15**	**12.14**	**10.96**
4	22.31	20.63	16	12.62	11.52
5	**24.57**	**22.83**	17	12.55	11.53
6	20.61	19.03	18	12.59	11.47
7	15.85	14.34	19	13.65	12.28
8	13.04	11.44	20	15.20	13.71
9	12.68	11.00	21	15.85	14.34
10	12.83	11.22	22	16.08	14.60
11	12.48	10.97	23	16.95	15.30

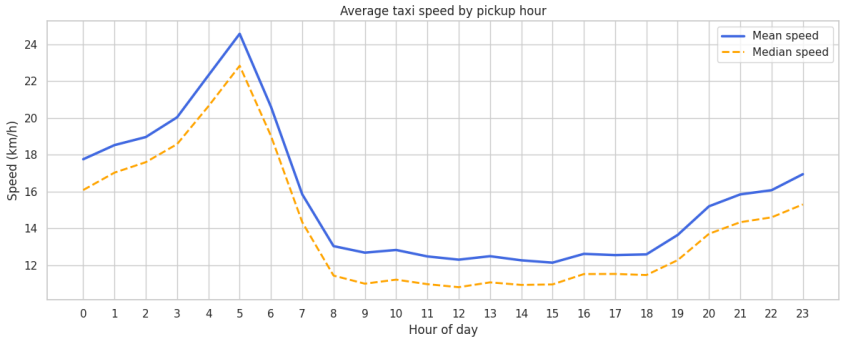


Figure 5: Mean and median taxi speed by pickup hour.

- **Fastest hour: 05:00, mean 24.57 km/h (median 22.83 km/h).** Roads are emptiest before the morning rush, and 04:00–06:00 is consistently the fastest block of the day.
- **Slowest hour: 15:00, mean 12.14 km/h (median 10.96 km/h).** From 08:00 to 18:00 the mean stays between 12.1 and 13.0 km/h, about half the early-morning speed. It recovers from 19:00 (13.65 km/h) as traffic thins.
- **Interpretation.** Speed follows road traffic rather than taxi demand alone. The whole working day is slow, not just the hours of highest taxi demand (18:00–19:00, Section 3.2). Section 5 finds the same pattern: trips to JFK take longest in the afternoon.

4 Location Clusters

Taxi demand in New York is concentrated in space and shifts through the day and week. This section maps that concentration with pickup heatmaps (Section 4.1) and then locates individual hotspots with DBSCAN for two driver-relevant periods (Section 4.2).

All analysis uses the 1,442,031 cleaned trips from Section 1. Pickups are further restricted to the map window

$$-74.05 \leq \text{longitude} \leq -73.75, \qquad 40.63 \leq \text{latitude} \leq 40.85,$$

which covers Manhattan, the western edge of Queens, northern and western Brooklyn, and the JFK and LaGuardia airports. Newark lies west of this window and is not shown.

4.1 Heatmaps

4.1.1 Choice of morning and evening bands

With roughly 1.4 million trips, a scatter plot saturates into a solid blob, so pickups are binned into grid cells and the cell counts are drawn as a heatmap. Trips are split by day type (Monday–Friday

against Saturday–Sunday) and by time of day. The time bands come from the hourly pickup profile in Figure 6.

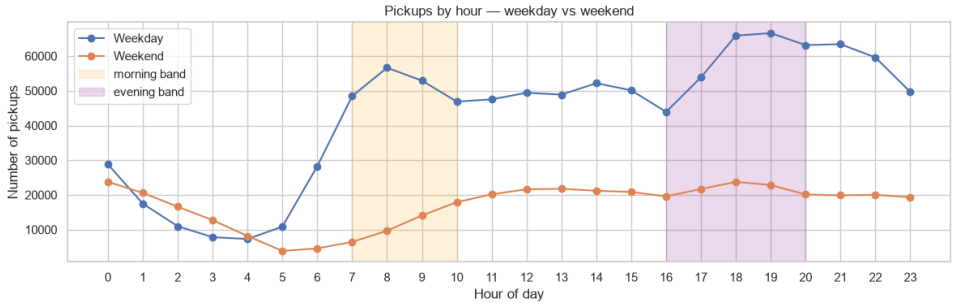


Figure 6: Total pickups by hour of day for weekdays and weekends. The shaded areas mark the morning band (07:00–09:59) and the evening band (17:00–19:59).

The weekday profile has two clear features:

- **Morning peak.** Pickups climb from about 11,000 at 05:00 to about 57,000 at 08:00, then drop to about 47,000 by 10:00 and stay between roughly 47,000 and 52,000 through the afternoon. The 08:00 maximum is the only pre-noon peak, so 07:00–09:59 captures the rise (07:00, about 48,000), the peak and the decline (09:00, about 53,000). A wider band starting at 06:00 would add an hour at about 28,000 that belongs to the ramp-up rather than the peak; a one-hour band at 08:00 would discard the shoulders that carry most of the volume.
- **Evening peak.** Pickups rise from a trough of about 44,000 at 16:00 to about 66,000 at 18:00 and 19:00, the highest values of the day. Demand stays elevated into the night (about 63,000 at 20:00 and 21:00, 60,000 at 22:00), so the evening is a broad plateau and not a narrow spike. The band 17:00–19:59 covers the steep rise and the two peak hours while staying clear of the late-evening hours, where the mix shifts towards leisure trips. This is the same window used in the second analysis for Thursday evening (17:00–20:00), which keeps the two parts of the section comparable.

Weekend volumes are much lower per hour and have no commute peak. Pickups are lowest at 05:00 (about 4,000), rise gradually through the morning, and sit between about 20,000 and 24,000 from 11:00 onwards, peaking weakly at 18:00. Weekend nights are comparatively busy: the 00:00 hour (about 24,000) exceeds the weekend morning band.

The same hours are used for both day types so that day type is the only variable that changes between panels. A weekend "morning" therefore is not a commute window; it measures where people are picked up in the same clock hours when commuters dominate on weekdays. The weekday–weekend gap in Figure 6 must also be read against the number of days: weekdays are five days and weekends two, so raw totals are not per-day rates. Volume per day of the week is in fact almost identical: each weekday has about 206,000 pickups over the six months, and so does each weekend day (Table 6), so the difference in hourly totals reflects the 5:2 day ratio plus a change in timing.

The timing change is visible in the share of each day type's pickups that fall in each band: The

Table 10: Pickups per group and share of all pickups on that day type (approximate; shares use day-of-week totals of the cleaned data).

	Weekday		Weekend	
Band	Pickups	Share	Pickups	Share
Morning (07:00–09:59)	157,911	15.3%	30,203	7.3%
Evening (17:00–19:59)	186,291	18.1%	68,272	16.6%

morning band holds twice as large a share of weekday trips as of weekend trips, while the evening band holds a similar share on both day types. Weekday mornings are therefore the most distinctive of the four groups, and the weekday and weekend evenings the most alike.

4.1.2 Grid construction

Each group's pickups are counted on one regular grid and every cell count is divided by the group's total, so each cell is the share of that group's pickups. This is needed because the groups differ in size by a factor of six (30,203 to 186,291 pickups); raw counts would make weekday groups look brighter everywhere without saying anything about where trips occur. All four panels share one logarithmic colour scale, since cell shares span about three orders of magnitude (from below 10^{-5} to above 10^{-2}) and a linear scale would show only Midtown.

Two grid versions are shown. The first (Figure 7) uses 150 bins in longitude and latitude, about 169 m by 163 m per cell, drawn on plain coordinates with the full range of cell shares on the colour scale. The cell size is close to the 150 m clustering radius used in Section 4.2, so the heatmaps and the hotspot analysis resolve the city at a similar scale. The second (Figure 8) uses a coarser 100-by-100 grid in Web Mercator coordinates (about 250 m cells) over an OpenStreetMap background, so cells can be matched to streets and neighbourhoods. Its colour scale runs from the 5th to the 99.5th percentile of non-empty cell shares, which prevents the few extreme cells (Midtown and the airports) from compressing the rest of the range; cells above the upper limit are drawn at maximum brightness.

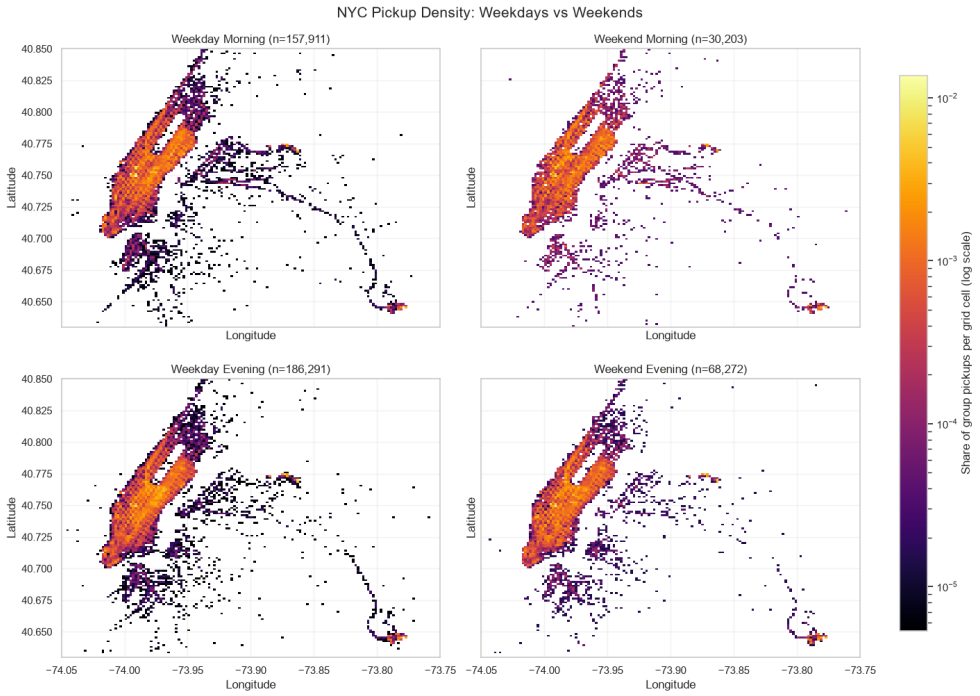


Figure 7: Pickup density for weekday and weekend mornings and evenings on a 150-by-150 grid (about 165 m cells). Cell values are the share of each group's pickups; the colour scale is logarithmic and common to all panels.

4.1.3 Findings

- **Demand is concentrated in Manhattan in every group.** The brightest cells form a band from the Financial District to roughly Harlem, strongest in Midtown and the Midtown-south corridor. Central Park appears as a dark hole in the middle of the band because there are few pickup points inside it. Outside Manhattan, pickups are sparse and form isolated cells.
- **The airports are the only bright cells outside Manhattan.** LaGuardia (about -73.87, 40.77) and JFK (about -73.78, 40.65) are visible in all four panels, and a thin trail of cells along the expressways linking JFK and LaGuardia to Manhattan is also visible. Airport pickups are the one non-Manhattan source of concentrated demand at every time studied.
- **Weekday mornings spread further into the outer boroughs.** The weekday morning panel shows the strongest secondary concentrations in downtown Brooklyn and in western Queens across the East River, consistent with commuters from residential areas heading

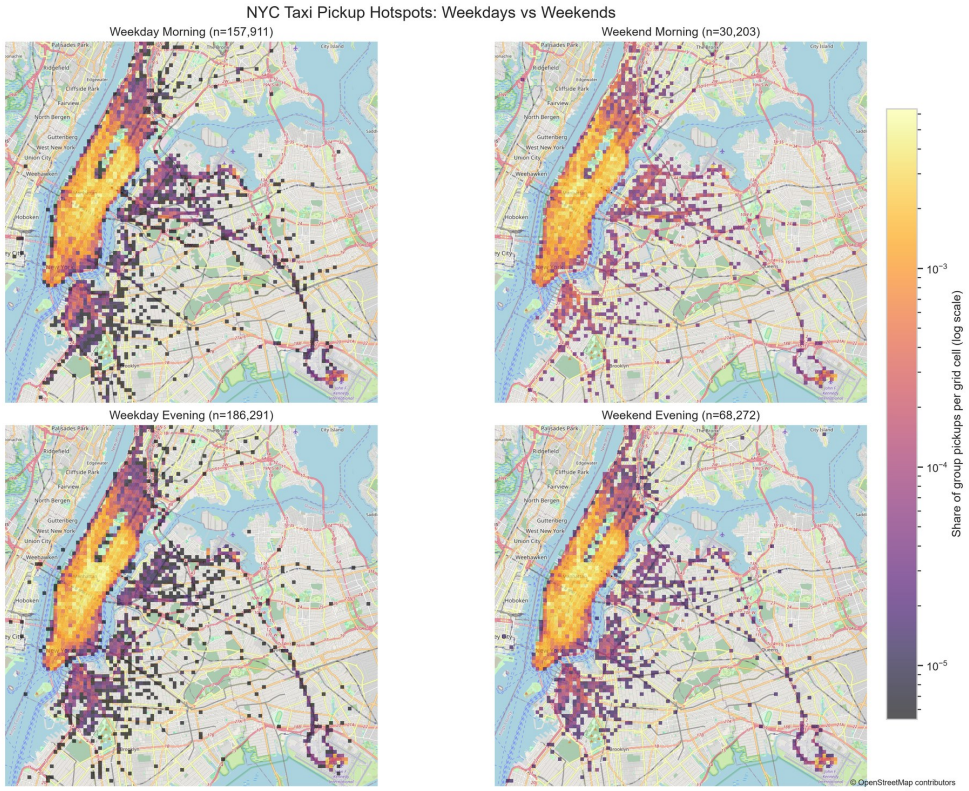


Figure 8: The same four groups on a 100-by-100 grid (about 250 m cells) over a street map, with a common logarithmic colour scale clipped at the 5th and 99.5th percentiles. Map tiles: © OpenStreetMap contributors.

into Manhattan. These areas are present but weaker on weekend mornings, where activity is dominated by Manhattan.

- **Evenings look similar across day types.** The weekday and weekend evening panels have a nearly identical Manhattan core, in line with the similar evening shares in Table 10. The weekend evening panel shows more cells in north-west Brooklyn, in the Williamsburg–Greenpoint direction, than the weekday evening panel.
- **Sample size matters for the sparse areas.** Scattered single cells in the weekend morning panel (30,203 pickups) are partly a sample-size effect: a cell with one pickup there has a larger share than a cell with one pickup in the 186,291-pickup weekday evening panel, so the weekend panels look more speckled even where the underlying distribution is similar. Differences in the Manhattan core are reliable; differences between individual outer cells are not.

Because every cell is a share within its group, the maps show where trips start relative to the rest of that group, not how many trips there are. They show that concentration exists and where, but not why a given area has more pickups.

4.2 Hotspot detection with DBSCAN

4.2.1 Periods and data

A driver planning an evening wants the places with the most pickups in a given hour. Two periods are examined:

- **Friday night:** Friday 23:00 to Saturday 02:00 (Friday 23:00–23:59 and Saturday 00:00–01:59).
- **Thursday evening:** Thursday 17:00 to 20:00 (17:00–19:59).

Pickups with missing timestamps or coordinates are dropped and the map window is applied. This leaves 34,868 pickups for Friday night and 38,255 for Thursday evening.

Table 11: Pickup records selected for DBSCAN.

Period	Selected pickups	Hourly windows (26 weeks)
Friday night (23:00–02:00)	34,868	78
Thursday evening (17:00–20:00)	38,255	78
Total	73,123	156

Each period spans three hours a week over the 26 weeks of the data, giving 78 separate hour-long windows per period and 156 in total. This corresponds to about 450 pickups per Friday-night hour and about 490 per Thursday-evening hour across the whole map window.

4.2.2 Method and parameter choice

DBSCAN [1] is used because the number of hotspots is not known beforehand. It grows clusters from points that have enough neighbours within a radius ε and labels isolated points as noise, so it needs no preset cluster count and does not force every pickup into a cluster, unlike k-means. Distances are great-circle (haversine) distances on coordinates converted to radians in latitude–longitude order, so ε is a true ground distance rather than a distance in degrees. Each hour-long window is clustered separately, since a hotspot is defined as a location with many pickups *within one hour*; pooling hours would merge places that are busy at different times with places that are busy at the same time.

Minimum samples. A hotspot requires at least 15 pickups within the neighbourhood in an hour, so `min_samples` $= 15$. This is one pickup every four minutes at a single location. It is a demanding threshold: a circle of radius 150 m has an area of 0.07 km^2, and 15 pickups represent about 3% of the roughly 450–490 pickups the entire map window produces in an average hour of these periods.

Radius. The radius is

$$\varepsilon = \frac{0.15}{6371.0088} \text{ radians } (= 150 \text{ m}),$$

where 6371.0088 km is the mean Earth radius. A radius of 150 m was chosen over the 50–100 m range because:

- Manhattan blocks are about 80 m between cross streets and 250–280 m between avenues, so 50 m cannot span even one block face; 150 m covers a street corner and its adjoining frontage, which is the area a driver can serve from one position.
- Urban-canyon GPS error between tall buildings is commonly tens of metres, so a very small radius splits one pickup spot into several fragments.
- The threshold is already strict. The detected clusters contain 15 to 17 pickups, meaning they only just exceed `min_samples`. Reducing ε reduces the number of neighbours each point can have, so smaller radii would find fewer, or no, clusters under the same threshold.

The cost of 150 m is a coarser location: a cluster centre can lie up to one block from the exact origin of its pickups. The results are conditional on this radius, and because the clusters sit at the minimum size, a change in either parameter would change the count. A comparison across several radii and thresholds would quantify that sensitivity; it has not been run here.

For each cluster the mean latitude, mean longitude and pickup count are recorded. Noise points are excluded.

4.2.3 Results

Six clusters are detected among the 156 hourly windows: four in Friday-night windows and two in Thursday-evening windows. Most windows (96%) contain no cluster at all, so a pickup concentration meeting the threshold is rare.

Three patterns stand out:

Table 12: Hourly DBSCAN cluster detections with approximate locations.

Period	Hour beginning	Latitude	Longitude	Pickups	Approximate area
Thursday evening	2016-01-28 19:00	40.770	−73.863	17	LaGuardia Airport
Friday night	2016-01-30 01:00	40.719	−73.990	16	Lower East Side
Friday night	2016-03-26 01:00	40.723	−73.987	16	East Village
Friday night	2016-04-16 01:00	40.740	−74.006	16	West Village
Friday night	2016-06-11 01:00	40.741	−74.006	15	West Village
Thursday evening	2016-01-07 18:00	40.749	−73.992	15	Midtown (Penn Station area)

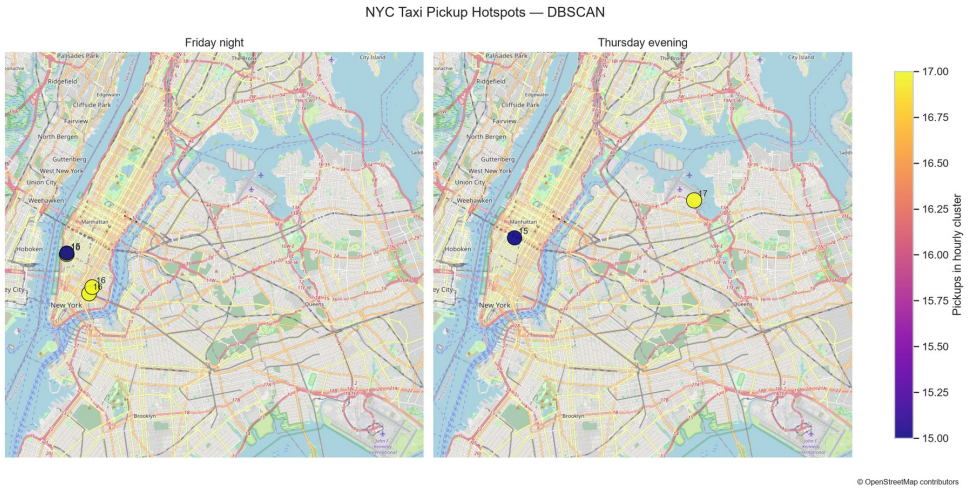


Figure 9: Hourly DBSCAN cluster centres for Friday night (left) and Thursday evening (right). Marker size and colour give the number of pickups in the cluster. Map tiles: © OpenStreetMap contributors.

- **Friday-night hotspots are nightlife areas at closing time.** All four Friday-night clusters fall in the 01:00–01:59 hour on a Saturday, and all lie in the downtown nightlife districts of the Lower East Side, East Village and West Village. None appears in the 23:00 or 00:00 hours, when pickups are spread more widely.
- **One location recurs.** Two Friday-night detections (16 and 15 pickups, April and June) are within about 100 m of each other near $(40.74, -74.006)$ and overlap on the map, so the six detections represent five distinct locations. This repetition on different dates suggests the West Village is a repeatable late-night hotspot, though two observations are too few to establish a pattern.
- **Thursday evening hotspots are transport hubs.** The largest cluster (17 pickups, 19:00 on 28 January) is at LaGuardia Airport, consistent with the airport cells in the heatmaps, and the other is in Midtown near Penn Station.

Answer. **Six** hourly clusters are found: four on Friday night and two on Thursday evening, covering five distinct locations. The result is small because the criterion is strict (15 pickups within 150 m in a single hour) and because the data are not dense enough to meet it outside a handful of places and times. Every cluster is within two pickups of the minimum, so the exact number depends on the parameters. The heatmaps show where pickups are generally concentrated, which is a wider region than the places that reach this hourly threshold.

4.3 Summary

Weekday pickups peak at 08:00 and 18:00–19:00, and the bands 07:00–09:59 and 17:00–19:59 capture those peaks. Normalising each group's grid by its own total shows that Manhattan dominates at all times, that LaGuardia and JFK are the only strong concentrations outside it, and that weekday mornings reach furthest into Brooklyn and Queens. The weekday morning band holds about twice

the share of the day's trips (15.3%) that the weekend morning band does (7.3%), while the evening bands are similar (18.1% and 16.6%).

DBSCAN with a 150 m radius and 15-pickup minimum finds six hourly hotspots among 156 windows: four downtown nightlife clusters during the 01:00 hour after Friday night, and two on Thursday evening at LaGuardia and in Midtown. These findings depend on the map window, time bands and the radius and threshold; the clusters are at the lower limit of what the criterion allows.

5 Airports: Travel Time Analysis

5.1 Selection of airport trips

The data does not identify the Empire State Building or the airports by name. It only provides GPS coordinates. We therefore calculated the distance between each taxi pickup and the ESB centre, and between each taxi drop-off and the centre of each airport. The Haversine formula was used to calculate distances in kilometres from the latitude and longitude values.

A taxi trip was included only when its **pickup was within 0.5 km of the ESB** and its **drop-off was within 2 km of either airport**. The analysis therefore considers travel *from the ESB to the airports*, not journeys in the opposite direction. We converted each selected trip's duration to minutes using

$$\text{Travel time (minutes)} = \frac{\text{Trip duration (seconds)}}{60}. \tag{1}$$

We calculated the overall mean travel time for each airport. To study time of day, we then grouped trips according to their *pickup hour* (00:00–23:00) and calculated the average duration and number of trips for each hour.

Additional output for Section 6. We also created an `airport_flags` table for the later clustering comparison in Section 6. Each of the 1,442,031 cleaned trips has an ID and indicators stating whether its pickup or drop-off was within 2 km of JFK or Newark. A trip is marked as airport-related when any one of these four indicators is true. The notebook identified **43,680 airport-related trips** (about **3.03%** of the cleaned data), with **no duplicate trip IDs**. This wider airport category is different from the ESB-to-airport subset used for the travel time calculations above.

5.2 Choice of radius

A radius is necessary because a taxi will not always stop at the exact centre coordinates. A passenger visiting the ESB might be picked up on a nearby street instead of directly outside the building. We chose a **0.5 km (500 m) ESB radius** to include these nearby streets while keeping the pickup area fairly small. A larger radius could include trips that started in other parts of Manhattan.

We chose a **2 km radius for JFK and Newark** because an airport covers a larger area than one building. Its terminals, access roads and drop-off areas may be some distance from the specified centre point. A very small radius could miss genuine airport drop-offs, a much larger one could include places outside the airport. Our chosen circles are approximations of airport areas, not exact boundaries.

To check whether our decisions affected the result, we performed a *radius sensitivity test*. First, we tested ESB radii of 0.3, 0.5 and 0.8 km while keeping the airport radius fixed at 2 km. We then tested airport radii of 1, 2 and 3 km while keeping the ESB radius fixed at 0.5 km. Table 13 gives the number of matched trips and the resulting average travel time for every test.

Increasing the radius usually included more trips, as expected. For example, changing the ESB radius from 0.3 km to 0.8 km increased JFK matches from 365 to 2,006 trips. However, the JFK average stayed between 49.26 and 49.67 minutes, and the Newark average stayed between 36.28 and 37.11 minutes. When the airport radius was changed from 1 to 3 km, JFK averages stayed between 48.79 and 49.44 minutes, and Newark averages between 36.26 and 36.43 minutes.

This means the *overall average travel times are not very sensitive to the radii tested*. We therefore kept 0.5 km for the ESB and 2 km for each airport. These settings give us a reasonable number of trips without making the location areas unnecessarily large. The test supports the stability of the overall means, but it does not prove that every matched trip is a genuine visit to that location.

Table 13: Radius sensitivity results. The final choice is an ESB radius of 0.5 km and an airport radius of 2 km.

Setting	JFK trips	JFK mean (min)	EWR trips	EWR mean (min)
ESB radius varied (airport radius = 2 km)				
ESB: 0.3 km	365	49.26	99	36.39
ESB: 0.5 km	**863**	**49.44**	**236**	**36.28**
ESB: 0.8 km	2,006	49.67	547	37.11
Airport radius varied (ESB radius = 0.5 km)				
Airport: 1 km	541	48.79	230	36.43
Airport: 2 km	**863**	**49.44**	**236**	**36.28**
Airport: 3 km	917	49.38	238	36.26

Using the selected radii, we found **863 trips from the ESB area to JFK** and **236 trips to Newark**. Table 14 shows the mean travel time for each route.

Table 14: Average duration of trips starting near the ESB and ending near each airport.

Destination	Number of trips	Mean travel time (minutes)
JFK Airport	863	49.44
Newark Airport	236	36.28

On average, journeys to **JFK took 13.16 minutes longer** than journeys to Newark. Part of this gap is distance: in a straight line, JFK is 21.1 km from the ESB and Newark 17.6 km. This comparison is based on the actual observed trips, so the results describe our selected 2016 taxi journeys rather than a guaranteed travel time for any future trip.

5.3 How travel time changes by time of day

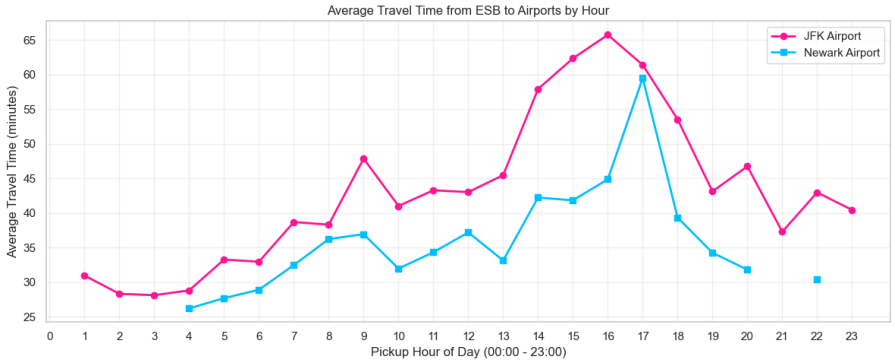


Figure 10: Average travel time from the ESB area to JFK and Newark by pickup hour. Times are in minutes, using the chosen location radii.

JFK: The shortest observed averages were in the early morning. For example, the mean at 04:00 was 28.83 minutes. Travel time generally increased as the day progressed, especially in the afternoon. It reached **65.75 minutes at 16:00**, which was the highest observed hourly mean for JFK. Between 14:00 and 17:00, the JFK averages were all above 57 minutes.

Newark: The same broad pattern appeared, although Newark was usually faster. At 04:00, the average was 26.22 minutes. Between 14:00 and 16:00, the averages were approximately 42–45 minutes. The highest observed Newark average was **59.50 minutes at 17:00**, but only four trips contributed to that value, so it is less reliable than an average based on many journeys.

A possible reason for longer afternoon journeys is heavier road traffic. Different route lengths and the bridges, tunnels or roads used may also contribute to the difference between the airports. However, the dataset does not contain direct traffic measurements or the routes followed. These are therefore *possible explanations*, not proven causes.

5.4 Number of trips and reliability of the hourly averages

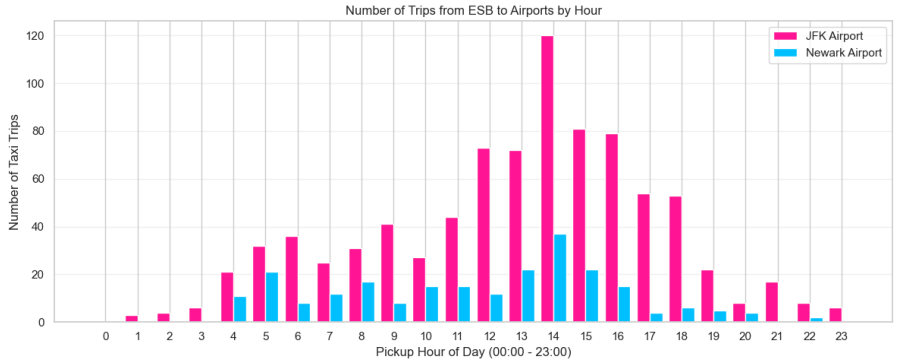


Figure 11: Number of ESB-to-airport trips by pickup hour. These are the sample sizes behind the hourly averages in Figure 10.

The largest number of trips to both airports occurred at 14:00: **120 JFK trips** and **37 Newark trips**. There were also **79 JFK trips at 16:00**, supporting the interpretation that the high JFK travel-time average at that hour is based on a substantial number of journeys. By contrast, Newark had only four trips at 17:00, so its sharp peak of 59.50 minutes should be interpreted cautiously. Newark also had no matching trips for several late-night hours, so no hourly average can be estimated for those hours from this sample.

6 Dimensionality Reduction

This section asks whether taxi trips fall into natural types. It uses the cleaned data from Section 1 (1,442,031 trips) and the airport flag from Section 5 (pickup or drop-off within 2 km of JFK or Newark). We reduce the data with PCA, t-SNE [2] and UMAP [3], and cluster with DBSCAN [1]. All random steps use seed 42.

6.1 Feature matrix

A random sample of 20,000 cleaned trips was drawn with `random_state=42`. It contains 598 airport trips (3.0%) and 28.5% weekend trips, close to the shares in the full cleaned data. Table 15 lists the eleven features.

Table 15: Features used for dimensionality reduction. All columns are standardised to zero mean and unit variance after the transform shown.

Group	Columns	Transform
Location	pickup/drop-off latitude and longitude	none
Trip size	distance (km), duration (s)	$\log(1+x)$
Trip pace	average speed (km/h)	none
Time of day	pickup hour h	$\sin(2\pi h/24)$, $\cos(2\pi h/24)$
Day of week	pickup day d (Mon $=0$)	$\sin(2\pi d/7)$, $\cos(2\pi d/7)$

Features. The four coordinates say where a trip happens. Distance, duration and speed say what kind of trip it is. Hour and day say when it happens. Together they cover the dimensions along which trip "types" (commutes, airport runs, late-night rides) would differ.

Encoding. Hour and day are cyclic: 23:00 is one hour from 00:00, and Sunday is next to Monday. Encoding them as an integer would place these neighbours at opposite ends of a line. The sine/cosine pair puts them on a circle instead.

Skew and scaling. Distance and duration are strongly right-skewed (skewness 2.79 and 2.46), so a few long airport trips would dominate a variance-based method. After $\log(1+x)$ their skewness falls to 0.96 and -0.20. Speed is only moderately skewed (1.46) and was left linear on purpose:

$\log(\text{speed}) = \log(\text{distance}) - \log(\text{duration}) + c$, so a logged speed would be an almost exact linear combination of the other two columns. The features also have very different units (degrees with a standard deviation near 0.03, seconds in the hundreds). PCA, t-SNE and UMAP all work with Euclidean distances, so unscaled data would be dominated by duration. Standard scaling gives each column the same weight. We chose it over min–max scaling because the coordinates contain a few far-away airport points that would compress everyone else into a narrow range.

The scaled matrix and the sample ids were saved as `q6_feature_matrix.csv` and `q6_sample_ids.csv`, so that every embedding in this section is computed on exactly the same trips.

6.2 PCA

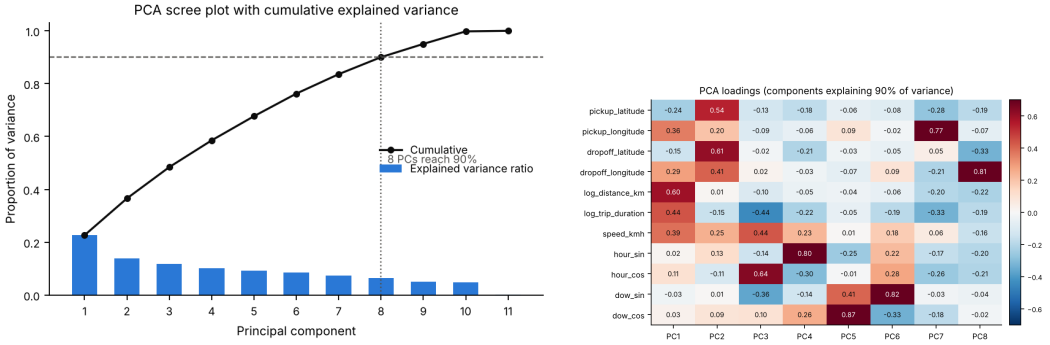


Figure 12: Left: explained variance per component and cumulative variance. Right: loadings of the eight components needed for 90% of the variance.

The variance is spread thinly (Figure 12). PC1 explains 22.7% and PC2 13.9%, and **eight of the eleven components are needed to reach 90%** (90.0%). The features are only weakly correlated, so a linear projection cannot compress them much. The last component explains just 0.2%: it is the near-dependency between log distance, log duration and speed.

Loadings. PC1 is a *trip length* axis. Log distance (0.60), log duration (0.44) and speed (0.39) load positively, together with pickup and drop-off longitude (0.36, 0.29). High PC1 therefore means long, fast trips heading east out of Manhattan. PC2 is a *north–south location* axis: drop-off and pickup latitude load at 0.61 and 0.54, separating uptown trips from downtown and Brooklyn trips. The time features barely appear in PC1 or PC2. Hour loads on PC3 and PC4 (cos: 0.64; sin: 0.80) and day of week on PC5 and PC6 (cos: 0.87; sin: 0.82).

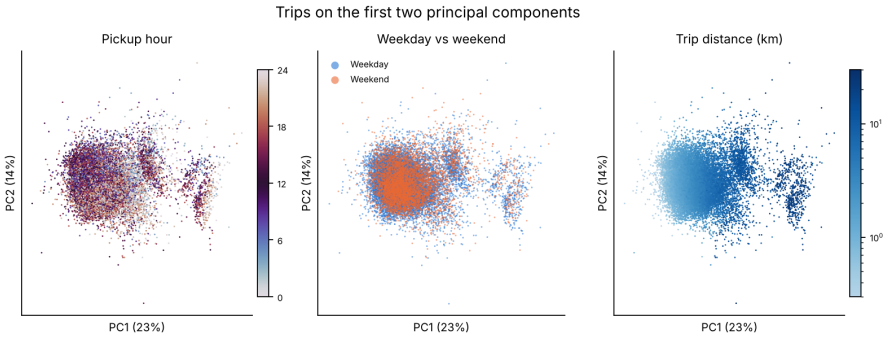


Figure 13: Trips on PC1 and PC2, coloured by pickup hour, weekday/weekend and trip distance (log colour scale).

Figure 13 confirms this reading. Distance increases smoothly from left to right, and the airport runs form a separate band on the far right. Hour and weekday/weekend show no pattern on these two axes, because that information sits on PC3–PC6. Apart from the airport band, the trips form one continuous cloud.

6.3 t-SNE

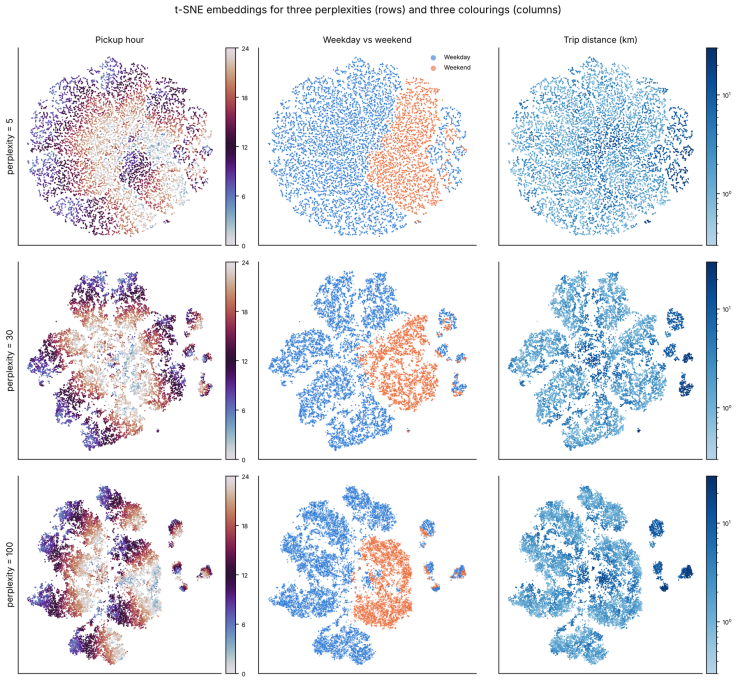


Figure 14: t-SNE embeddings (`init="pca"`) for perplexity 5, 30 and 100 (rows), coloured by pickup hour, weekday/weekend and distance (columns).

In t-SNE, perplexity sets the effective number of neighbours each point tries to keep close [2] (Figure 14). At **perplexity 5** the embedding is a single disc of short strands. Neighbourhoods are tiny, so only very local similarity survives, and the only visible split is between weekday and weekend trips. At **perplexity 30** the disc breaks into lobes, and small separate islands of long trips appear on the right. At **perplexity 100** the lobes become compact, weekday and weekend occupy clearly different regions, and the long-trip islands (airport and LaGuardia runs) are cleanly separated. Within each lobe the hour colouring runs as a smooth gradient from morning to night, so t-SNE arranges time of day *inside* the groups rather than splitting on it.

Larger perplexity therefore gives coarser, more global structure but costs more time (86 s, 112 s and 161 s). Measured on a fixed subset of 3,000 trips, trustworthiness at $k = 15$ rises from 0.943 to 0.978 and the correlation of pairwise distances with the original space from 0.295 to 0.357 (Table 16).

6.4 UMAP

In UMAP [3], `n_neighbors` plays the same role as perplexity (Figure 15). With 5 neighbours the map breaks into many thin fragments. With 50 it settles into seven large lobes plus a few small islands, and the global distance correlation improves from 0.319 to 0.461 (at `min_dist` $= 0$).

`min_dist` only controls how tightly points are packed. At 0.0 the lobes are dense and separated by clear gaps, which suits density-based clustering. At 0.5 the points spread out, the gaps close and the lobes merge into a single flower shape. The structure underneath is the same, but it is harder to see where one group ends: trustworthiness drops from 0.964 to 0.928 at 50 neighbours.

In every setting the long trips (dark blue) separate from the rest. The hour and weekend colourings show that the seven lobes are days of the week, with hour as a gradient along each lobe. Section 6.6 returns to this. UMAP was much faster than t-SNE (6–24 s per run; the first run also compiles the library and was the slowest).

Best embeddings. We chose the setting with the highest mean of the two scores, so that neither local nor global structure is sacrificed: t-SNE with perplexity 100, and UMAP with `n_neighbors`$= 50$, `min_dist`$= 0.0$. These are used in the rest of the section.

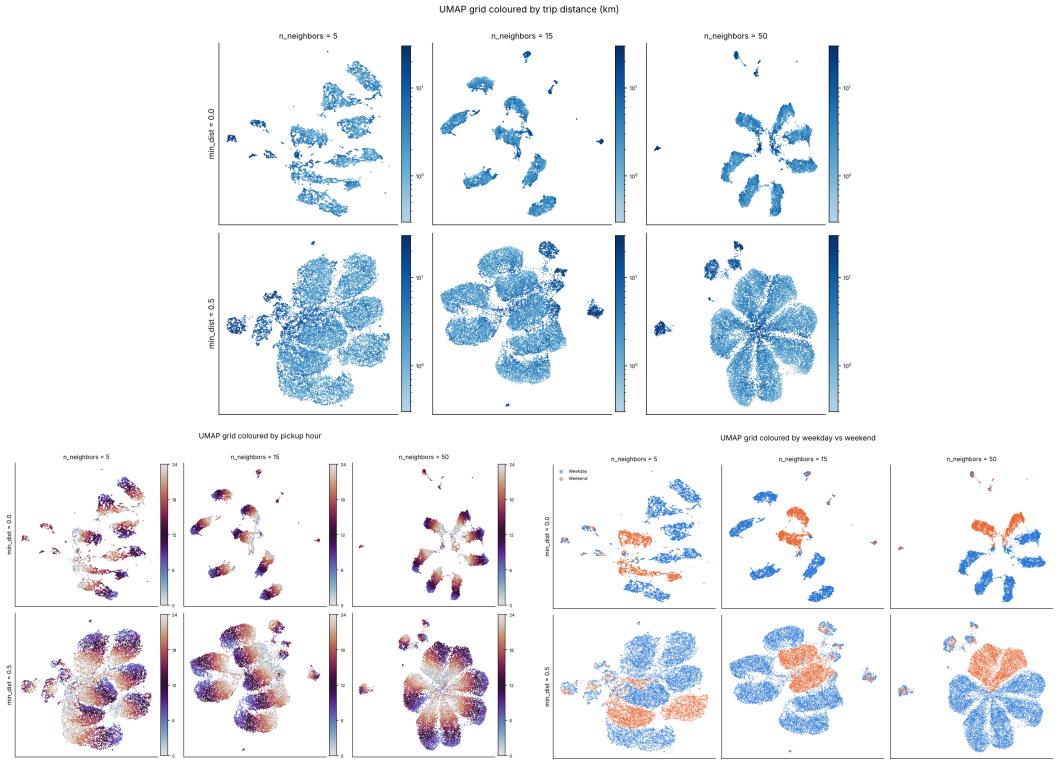


Figure 15: UMAP for `n_neighbors` $\in \{5, 15, 50\}$ (columns of each grid) and `min_dist` $\in \{0.0, 0.5\}$ (rows), coloured by trip distance (top), pickup hour (bottom left) and weekday/weekend (bottom right).

Table 16: Structure preserved by each 2D embedding, measured on a fixed random subset of 3,000 trips. Trustworthiness ($k = 15$) measures local structure; Spearman ρ between pairwise distances in the scaled feature space and in the embedding measures global structure. The best setting per method (highest mean) is in bold.

Method	Setting	Trustworthiness	Spearman ρ	Mean
PCA	PC1–PC2	0.758	0.766	0.762
t-SNE	perplexity 5	0.943	0.295	0.619
t-SNE	perplexity 30	0.971	0.313	0.642
t-SNE	**perplexity 100**	**0.978**	**0.357**	**0.667**
UMAP	n=5, d=0.0	0.953	0.319	0.636
UMAP	n=15, d=0.0	0.960	0.368	0.664
UMAP	**n=50, d=0.0**	**0.964**	**0.461**	**0.713**
UMAP	n=5, d=0.5	0.916	0.294	0.605
UMAP	n=15, d=0.5	0.929	0.353	0.641
UMAP	n=50, d=0.5	0.928	0.387	0.658

6.5 Density-based clustering

Method and parameters. DBSCAN [1] groups points that have at least `min_samples` neighbours within a radius `eps` and labels the rest as noise. It was run on the eight principal components that explain 90% of the variance, and on the best t-SNE and UMAP embeddings. `min_samples` was fixed first. For PCA we used the usual heuristic of twice the dimension, $2 \times 8 = 16$. For the 2D embeddings that heuristic gives 4, which lets DBSCAN cut the dense embeddings into dozens of tiny clusters. We instead required 50 trips (0.25% of the sample), the smallest group we would call a recurring trip type. `eps` was then read off the k-distance curve (distance to the kth neighbour, $k =$ `min_samples`), at the knee: the point furthest from the straight line joining the ends of the sorted curve (Figure 16).

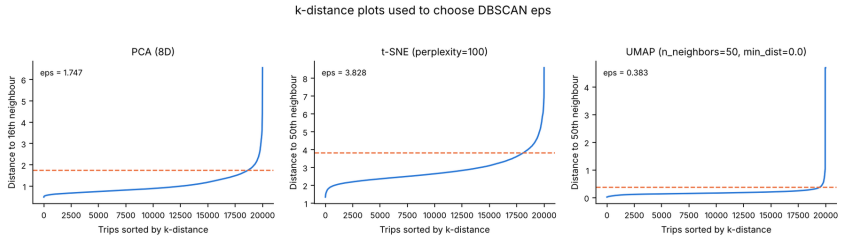


Figure 16: Sorted k-distance curves. The dashed line is the knee used as `eps`.

Table 17: DBSCAN settings and results in each reduced space.

Space	`min_samples`	`eps`	Clusters	Noise
PCA (8 components)	16	1.747	3	2.2%
t-SNE (perplexity 100)	50	3.828	10	1.1%
UMAP (n=50, d=0.0)	50	0.383	13	0.8%

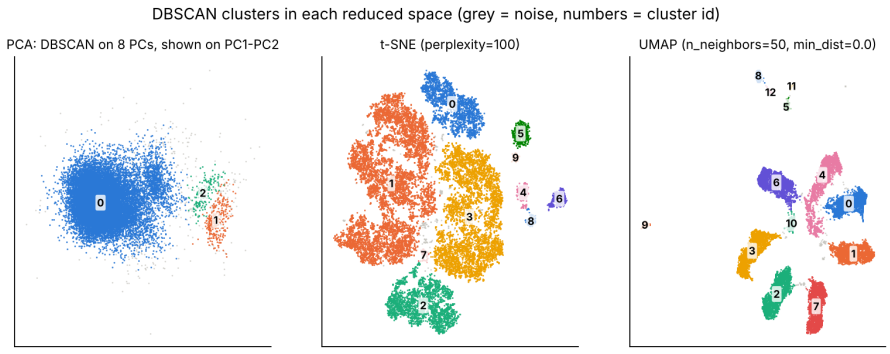


Figure 17: DBSCAN clusters. PCA clusters are found in eight dimensions and drawn on PC1–PC2. Grey points are noise; numbers are cluster ids, matching Table 18.

Profiles and names. Each cluster was profiled on the original, unscaled features (Table 18). The airport columns use the flag from Section 5, split by direction. Because several clusters were clearly LaGuardia trips, we also flagged pickups and drop-offs within 2 km of LaGuardia (40.7769, −73.8740). This flag only describes clusters; it is not used as a feature or a filter. Names follow fixed rules: airport direction first, then day of week if one day makes up more than 80% of the cluster, otherwise time of day, weekday/weekend, length and area.

The three spaces tell one consistent story about the long trips and three different stories about everything else.

- **PCA** finds only three groups. One holds 95% of trips (Manhattan trips at all times: median 2.0 km and 10.6 min). The other two are *JFK/Newark pickups to Manhattan* (294 trips) and *Manhattan rides to JFK/Newark* (171 trips), both about 20 km and 40–45 min.

Table 18: Cluster profiles (original units; distance, duration and speed are medians). Hour is the circular mean pickup hour, used instead of the median because a plain median treats 23:00 and 00:00 as 23 hours apart. Wknd is the weekend share; Apt is the share of airport trips (Section 5 flag). Clusters are listed by size.

Space	Name (id)	Trips	Dist (km)	Dur (min)	Speed	Hour	Wknd	Apt
PCA	evening all-week medium Manhattan trips (0)	19,091	2.0	10.6	12.6	16:36	0.28	0.00
	JFK/Newark pickups to Manhattan (1)	294	20.3	43.6	26.9	17:57	0.28	0.98
	Manhattan rides to JFK/Newark (2)	171	20.4	41.3	29.2	11:58	0.30	0.90
t-SNE	evening weekday medium Manhattan trips (1)	7,958	1.9	10.7	11.6	16:12	0.01	0.00
	evening weekend medium Manhattan trips (3)	5,607	2.2	10.1	14.3	18:36	0.93	0.00
	Friday Manhattan trips (2)	2,586	1.8	10.2	11.0	16:36	0.00	0.00
	Monday Manhattan trips (0)	2,105	1.7	9.1	12.1	15:26	0.00	0.00
	LaGuardia pickups to Manhattan (5)	513	9.8	30.7	20.2	16:35	0.25	0.00
	JFK/Newark pickups to Manhattan (6)	411	19.9	39.1	27.0	18:01	0.28	0.97
	Manhattan rides to LaGuardia (4)	292	9.8	26.9	22.3	11:02	0.28	0.00
	Manhattan rides to JFK/Newark (8)	197	20.3	39.9	30.5	12:05	0.29	0.84
	late-night Friday Manhattan trips (7)	62	1.7	3.6	27.9	04:58	0.00	0.00
	LaGuardia pickups to Queens (9)	50	3.2	13.3	14.6	12:40	0.24	0.00
UMAP	Sunday Manhattan trips (4)	2,896	2.4	10.5	15.0	19:36	0.86	0.00
	Friday Manhattan trips (3)	2,728	1.8	10.1	11.3	16:44	0.00	0.00
	Saturday Manhattan trips (6)	2,672	2.0	9.8	12.8	17:58	0.99	0.00
	Thursday Manhattan trips (2)	2,641	1.8	10.4	11.2	15:44	0.00	0.00
	Wednesday Manhattan trips (7)	2,497	1.8	10.1	11.3	15:35	0.00	0.00
	Tuesday Manhattan trips (1)	2,393	1.8	10.1	11.4	15:11	0.00	0.00
	Monday Manhattan trips (0)	2,256	1.8	9.3	12.5	15:23	0.00	0.00
	LaGuardia pickups to Manhattan (8)	516	9.8	30.3	20.2	16:34	0.25	0.00
	JFK/Newark pickups to Manhattan (9)	411	19.9	39.1	27.0	18:01	0.28	0.97
	Manhattan rides to LaGuardia (5)	300	9.9	26.7	22.7	11:10	0.28	0.00
	late-night all-week long Manhattan trips (10)	294	10.5	20.2	30.5	00:49	0.43	0.00
	Manhattan rides to JFK/Newark (11)	188	20.3	40.5	30.4	11:49	0.30	0.88
	LaGuardia pickups to Queens (12)	53	3.1	12.8	15.0	12:13	0.24	0.00

- **t-SNE** finds the same two airport runs (411 and 197 trips), adds two LaGuardia runs (513 into Manhattan, 292 out to LaGuardia) and a small *LaGuardia pickups to Queens* group of short local trips. It splits the large Manhattan mass into weekday (7,958) and weekend (5,607) groups, with separate Friday and Monday lobes. It also finds a 62-trip *late-night Friday Manhattan trips* cluster: very short (3.6 min), fast (28 km/h) trips with a mean pickup time near 05:00.
- **UMAP** finds all of the airport and LaGuardia runs again, plus a *late-night long trip* group: 294 trips with a mean pickup hour around 01:00, 10 km and 30 km/h. It splits the Manhattan mass into seven clusters, **one for each day of the week** (each over 99% one day, except Sunday at 83%).

The pickup maps (Figures 18 and 19) make the difference visible. The seven day clusters cover exactly the same part of Manhattan, while the airport and LaGuardia clusters pick up at the airport or drop off in Midtown. This matches Section 4, where LaGuardia and JFK are the only strong pickup concentrations outside Manhattan in the heatmaps, and LaGuardia hosts the largest Thursday-evening hotspot.

6.6 Comparison

Groups that appear in all three methods. Airport trips form their own group in every method (Figure 20, top row). Of the 598 airport trips in the sample, 66.6% fall in a single JFK/Newark-to-Manhattan cluster in both t-SNE and UMAP, and that cluster is 96.8% airport trips. PCA also isolates them (98.3% pure), but it sends 26.1% of the airport trips to noise, against 5.5% for t-SNE and UMAP. Direction matters: trips *from* the airport and trips *to* the airport form separate clusters in every space, because pickup and drop-off coordinates are separate features. These long-trip groups, together with LaGuardia runs in t-SNE and UMAP, are the only structure all three methods agree on.

What PCA reveals and misses. PCA gives axes with a meaning (trip length and north–south location) and keeps global distances best ($\rho = 0.766$). It shows that most trips form one continuum rather than separate types. Because it is linear and keeps only two of the eight important directions in the plot, it misses everything carried by time: hour and day are invisible on PC1–PC2, and in eight dimensions DBSCAN still sees a single main cloud. Its local structure is the weakest (trustworthiness 0.758). The non-linear methods recover the time structure that PCA spreads across PC3–PC6.

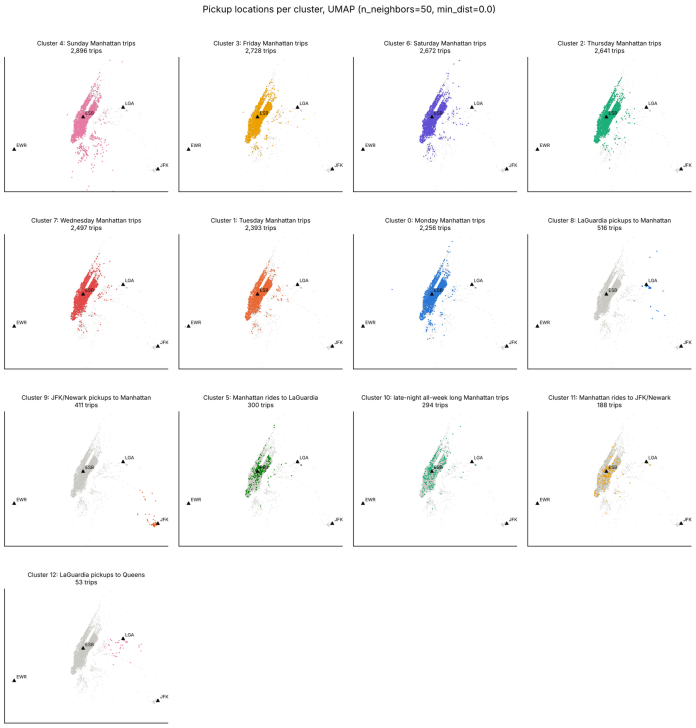


Figure 18: Pickup locations of each UMAP cluster (colour) against all sampled pickups (grey).

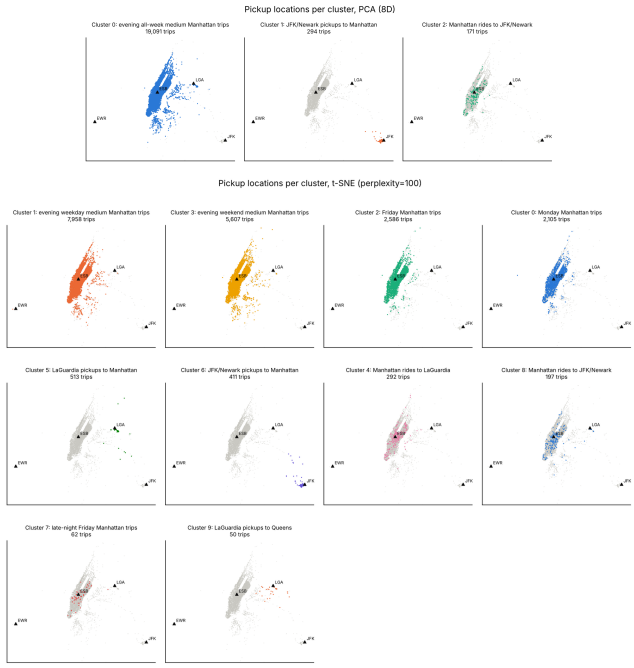


Figure 19: Pickup locations of each PCA cluster (top) and each t-SNE cluster (bottom).

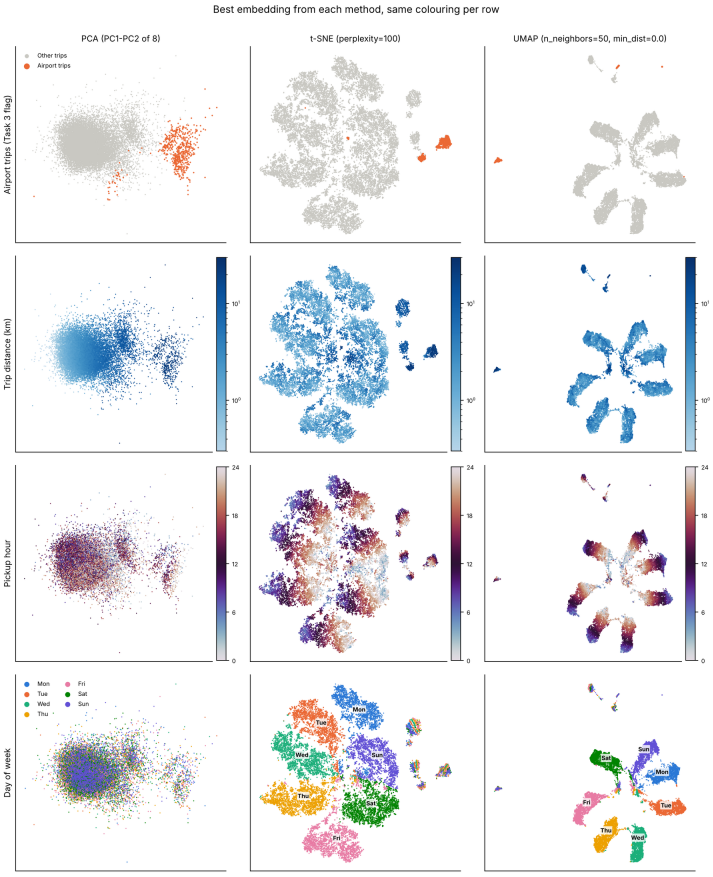


Figure 20: Best embedding per method (columns) with the same colouring in each row: Section 5 airport flag, distance, pickup hour and day of week.

Local and global structure. t-SNE and UMAP preserve neighbourhoods well (trustworthiness 0.978 and 0.964) but global distances poorly ($\rho = 0.357$ and 0.461). Within a lobe, nearby points really are similar trips. Distances *between* lobes, the empty space separating them and the area each lobe covers cannot be read as similarity or size. t-SNE in particular equalises density, so a small, tight group of airport runs and a large diffuse group of Manhattan trips can end up looking alike. UMAP keeps somewhat more of the global layout (the airport islands sit far from Manhattan, as they should), but its gaps are still not to scale. Only in PCA are distances and spreads directly interpretable.

The day-of-week structure. The bottom row of Figure 20 shows that the seven UMAP lobes, and most t-SNE lobes, are days of the week. Day of week takes only seven values, so after the cyclic encoding every day sits at its own fixed point on a circle. Adjacent days are 1.22 apart in scaled units, more than the median distance from a trip to its nearest neighbour (0.59) or even its 15th-nearest neighbour (1.08). Each day therefore becomes a separate shelf of the data. Methods built on nearest neighbours turn these shelves into islands. The agreement between clusters and day of week, measured by ARI, is 0.83 for UMAP, 0.45 for t-SNE and 0.00 for PCA. As a check we reran the chosen UMAP without the two day-of-week columns (Figure 21). The lobes disappear, and DBSCAN with the same rule finds six clusters: one main Manhattan mass (18,266 trips), the four airport and LaGuardia runs, and a late-night group of long Manhattan-to-Queens trips (148 trips, mean pickup hour about 00:00). The ARI against the original UMAP clusters is only 0.045.

Runtime and sensitivity. On the same 2-core machine, PCA takes a fraction of a second, has no tuning parameters and gives the same answer for any seed. UMAP takes 6–24 s per run and t-SNE 86–161 s, growing with perplexity. Both non-linear methods change markedly with their main parameter (Figures 14 and 15). To test the seed, we reran the chosen settings with seeds 0 and 7 and compared each run with seed 42 (Table 19). With PCA initialisation, t-SNE returned exactly the

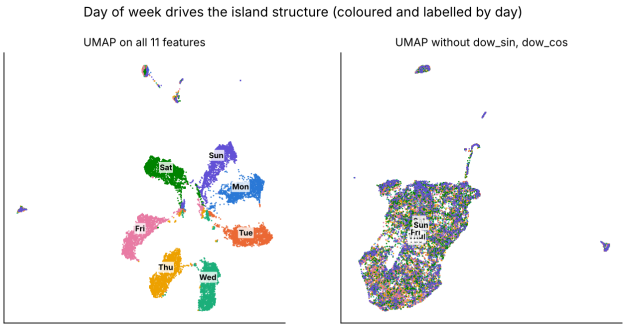


Figure 21: The chosen UMAP on all eleven features (left) and without `dow_sin`/`dow_cos` (right), coloured by day of week.

same embedding for all three seeds (Procrustes disparity 0, identical neighbours). In scikit-learn the seed only enters through the initial layout, and PCA initialisation fixes that layout. With random initialisation the layout changed (disparity 0.10–0.13, and 74–76% of each trip's 15 neighbours kept), and DBSCAN found between 8 and 12 clusters (ARI 0.61–0.76 against seed 42). UMAP changed most at the level of individual neighbours: only 41–43% of neighbours were kept, with disparity 0.11–0.33. Its clusters were fairly stable, though (9 to 11 clusters against 13, ARI 0.71–0.80). In every run the airport and LaGuardia islands reappear, while some of the day lobes merge or split. PCA has no seed dependence.

Table 19: Seed sensitivity of the chosen settings, comparing seeds 0 and 7 with seed 42. Procrustes disparity is 0 for identical layouts up to rotation and scale. kNN overlap is the mean share of each trip's 15 nearest embedding neighbours that are the same across seeds. ARI compares DBSCAN clusters found with the same rule.

Method	Seed	Procrustes	kNN overlap	Clusters	ARI vs seed 42
t-SNE	0	0.000	1.000	10	1.000
t-SNE	7	0.000	1.000	10	1.000
t-SNE (random init)	0	0.102	0.757	12	0.612
t-SNE (random init)	7	0.132	0.735	8	0.761
UMAP	0	0.111	0.433	11	0.800
UMAP	7	0.328	0.409	9	0.710

Agreement between methods. Table 20 gives the adjusted Rand index between each pair of clusterings. We handle noise in two ways: treating noise as one extra label (all points), and keeping only trips that both methods assign to a cluster.

Table 20: Adjusted Rand index between cluster labels.

Pair	ARI, noise as a label	ARI, clustered in both	Share clustered in both
PCA vs t-SNE	0.054	0.037	97.0%
PCA vs UMAP	0.021	0.014	97.2%
t-SNE vs UMAP	0.480	0.490	98.3%

Agreement is low with PCA and moderate between t-SNE and UMAP. The two ways of handling noise give almost the same values, because all methods label very few trips as noise. ARI is dominated by how the large Manhattan mass is divided: PCA keeps it whole, t-SNE splits it mainly into weekday and weekend, and UMAP into seven days. The small airport and LaGuardia clusters, which match closely across methods, barely move the index. A low ARI therefore does not mean the methods disagree about everything. It means they disagree about the part of the data that has no clear clusters.

Can t-SNE and UMAP clusters be trusted? Partly. We checked each clustering in the original 11-dimensional space by asking how many of a trip's 15 nearest neighbours there share its cluster. The answer is 0.970 for t-SNE and 0.957 for UMAP, against chance levels of 0.272 and 0.121. So the clusters are not invented by the projection: they are coherent regions of the feature space. Coherent is not the same as natural, though. The day-of-week clusters are coherent only because we put day

of week into the feature matrix, and they disappear once those two columns are removed. Clusters found in an embedding are hypotheses that must be checked against the original features and against other methods. Here the airport and LaGuardia runs pass every check, while the day-of-week clusters reflect our encoding rather than the trips.

Recommendation. For exploring this dataset we recommend **UMAP** (`n_neighbors`$= 50$, `min_dist`$= 0.0$), used together with PCA. UMAP had the best balance of local and global structure in Table 16, recovered every group the other methods found, and ran many times faster than t-SNE, which makes it practical to try several feature sets and parameters. PCA should sit beside it because its axes can be interpreted and its distances can be trusted, which keeps us from reading too much into UMAP's islands. t-SNE produced similar groups but was the slowest and is no more interpretable. Whatever the method, clusters should be named and checked from profiles in the original units, as in Table 18. The analysis also suggests treating day of week as a colouring or a filter rather than a feature when the goal is to find trip types.

Conclusion

Data (Sections 1–2). Five physically motivated filters removed 1.14% of the 1,458,644 trips, leaving 1,442,031 for every later analysis. Haversine distance, average speed and the day, hour and weekend features were added.

When (Section 3). Friday is the busiest day and Monday the quietest. Weekdays peak in the evening commute (18:00–19:00) with a smaller 08:00 peak; weekends have no morning peak and are busiest around midnight. Of the five holidays, only Memorial Day (-23%) and Valentine's Day ($+13\%$) change the daily total noticeably, and the two Monday holidays lose much of the morning commute. Average speed is lowest throughout the working day (12.1–13.0 km/h from 08:00 to 18:00) and highest at 05:00 (24.6 km/h).

Where (Section 4). Pickups are concentrated in Manhattan at all times, and LaGuardia and JFK are the only strong concentrations outside it. DBSCAN with a 150 m radius and 15 pickups per hour finds six hotspots: four downtown nightlife spots in the 01:00 hour after Friday night, and LaGuardia and Penn Station on Thursday evenings.

Airports (Section 5). Trips from the Empire State Building take 49.4 minutes on average to JFK (863 trips) and 36.3 minutes to Newark (236 trips). Travel time is shortest in the early morning and longest in the afternoon, peaking at 65.8 minutes to JFK at 16:00, the same daytime slowdown seen in Section 3.5.

Trip types (Section 6). PCA needs 8 of 11 components for 90% of the variance, and most trips form one continuum. The only groups that PCA, t-SNE and UMAP all recover are airport runs, split by direction, with LaGuardia runs added by the non-linear methods. These match the airport hotspots of Section 4. The island structure that t-SNE and UMAP show elsewhere mostly reflects the day-of-week encoding rather than real trip types. We recommend UMAP used together with PCA, with every cluster checked against the original features.

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